DISTECH CONTROLS

Advanced Support Tools

Eclypse Backup Dissector

User Guide

Document Information	
Product	Eclypse Backup Dissector 1.0.0.exe
Version	1.0.0
Audience	Distech Controls Support Engineers
Classification	Internal / Support Use

1. Overview

The **Eclypse Backup Dissector** is a Windows desktop utility designed for Distech Controls support engineers. It allows you to open, inspect, and extract the contents of Eclypse controller backup ZIP archives without modifying the original file.

The tool displays a structured inventory of expected backup artifacts, highlights what is present or missing, renders rich HTML configuration reports for key JSON files directly inside the application, and can repackage designer projects and dashboard files into deployment-ready ZIP archives.

1.1 Key Capabilities

- **Non-destructive inspection —** the source ZIP is opened read-only at all times.
- **Artifact scanning —** instantly identifies which critical files are present or absent.
- **Inline rich reports —** bacnet.json, network.json, modbus.json, security.json, and iot.json render as tabbed HTML reports inside the application window.
- **Full extraction —** unpacks the entire ZIP to a local workspace.
- **Re-packaging —** creates DesignerProjects.zip and Dashboard.zip from extracted content.
- **Diagnostic logging —** every action is logged to a timestamped file in the workspace.
- **Zip Slip protection —** malicious path traversal entries in ZIP files are detected and refused.

2. System Requirements

Requirement	Detail
Operating System	Windows 10 or Windows 11 (64-bit)
Runtime	No installation required — self-contained executable
Screen Resolution	Minimum 1280 × 720; maximised window recommended
Disk Space	Workspace folder: size depends on backup contents (typically 50 MB – 500 MB)
Permissions	Standard user account; no administrator rights required

3. Getting Started

3.1 Launching the Application

Double-click `Eclypse Backup Dissector 1.0.0.exe` to launch the tool. The application opens maximised by default. A temporary workspace folder is created automatically in your system's temp directory.

3.2 Application Layout

The interface is divided into four main areas:

Area	Purpose
Top bar — ZIP & Workspace	Enter or browse for the source ZIP file and set the local workspace folder.
Action toolbar	Buttons for extracting, previewing, saving, and re-zipping artifacts.
Artifact grid (left pane)	Lists all expected backup artifacts with their status, path inside the ZIP, and extracted path.
Preview pane (right pane)	Displays rich HTML reports or plain-text previews of the selected artifact.
Diagnostic log (bottom)	Expandable log panel showing timestamped application events.

4. Recommended Workflow

Step 1 — Select the ZIP file

Click **Browse...** next to the ZIP file field and navigate to the Eclypse backup archive, or type the full path directly into the text box.

Step 2 — Load / Scan

Click **Load ZIP / Scan**. The tool opens the ZIP read-only, enumerates its contents, and populates the artifact grid. Each expected file or folder tree is listed with one of two statuses:

Status	Meaning
Present (green)	The file or folder tree was found inside the ZIP.
Missing (red)	The expected artifact was not found in the ZIP.

Step 3 — Preview an artifact

Select any row in the artifact grid and click **Preview Selected**. The preview pane on the right updates immediately:

- **JSON configuration files** (bacnet.json, network.json, modbus.json, security.json, iot.json) — rendered as tabbed HTML reports with collapsible sections, colour-coded badges, and a Raw JSON tab for the original source.
- **Plain text files** — displayed in a monospace viewer.
- **Binary files** — a notice is shown; use Extract then Open Selected to inspect.

> **Tip — Preview without extracting**
> All previews read directly from the ZIP. You do not need to run Extract before previewing.

Step 4 — Extract (optional)

Click **Extract** to unpack the entire ZIP to the `ExtractedFiles` subfolder of the workspace. A confirmation dialog is shown before proceeding. Extraction is required before using **Open Selected**, **Save As...**, or the re-packaging buttons.

Step 5 — Re-package (optional)

After extraction, use the two orange buttons to create deployment-ready archives:

- **Create DesignerProjects.zip** — packages the `designer/proj/` tree into a ZIP suitable for importing into EC-BOS Designer.
- **Create Dashboard.zip** — packages the `files/dashboards/` tree into a ZIP for dashboard restoration.

Both output ZIPs are saved to the `Zips` subfolder of the workspace. Click **Open Output Folder** to open that folder in Windows Explorer.

5. Button Reference

Button	Description
Browse...	Opens a file picker to select the source Eclypse backup ZIP.
Load ZIP / Scan	Scans the selected ZIP and populates the artifact grid. The ZIP is never modified.
Change... (Workspace)	Opens a folder picker to select a different workspace directory.
Extract	Unpacks the entire ZIP to the ExtractedFiles subfolder. Prompts for confirmation first.
Preview Selected	Renders the selected artifact in the preview pane. Works directly from the ZIP — no extraction needed.
Open Selected	Opens the extracted file with its default Windows application. Requires extraction.
Save As...	Copies the selected extracted file to a location of your choice.
Create DesignerProjects.zip	Repackages the extracted designer project tree into a deployment-ready ZIP.
Create Dashboard.zip	Repackages the extracted dashboards folder into a deployment-ready ZIP.
Open Output Folder	Opens the Zips subfolder of the workspace in Windows Explorer.
Reset / Clear	Clears the grid, resets the ZIP path, and creates a fresh workspace. Does not delete any files.
Save Log...	Exports the diagnostic log to a .log or .txt file.
Clear Log	Clears the in-memory log buffer and the log display.

6. Artifact Reference

The following artifacts are scanned in every backup ZIP:

Artifact	Category / Notes
bacnet.json	BACnet configuration — device info, objects, schedules, network ports, remote devices.
Project.gfx	Graphical project file for EC-BOS Designer.
network.json	Network interface configuration — IP settings, DHCP, VLANs, routing, spanning tree.
security.json	Security configuration — user accounts, roles, lockout policies, authentication settings.
modbus.json	Modbus configuration — server settings, client devices, and point definitions.
iot.json	IoT connection configuration — cloud provider connections and secrets.
designer/proj/	Designer project tree (folder). Contains all project source files.
files/dashboards/	Dashboard files tree (folder). Contains all dashboard JSON and assets.
DesignerProjects.zip	Output artifact — created by the Create DesignerProjects.zip button.
Dashboard.zip	Output artifact — created by the Create Dashboard.zip button.

7. Rich HTML Reports

Selecting and previewing any of the five key JSON files produces a tabbed HTML report rendered directly inside the application. A copy of the report is also written to the workspace root as `report_<name>.html` for external reference.

7.1 bacnet.json Report Tabs

Tab	Content
Device Info	Local BACnet device instance, object name, database revision, and segmentation support.
Analog Inputs	All AI object instances with names and engineering units.
Analog Values	All AV instances with units, default values, alarm limits, and notification class.
Binary Inputs	All BI instances with object names.
Binary Values	All BV instances with active/inactive text and default values.
Multi-State Values	All MSV instances with state text enumeration and defaults.
Schedules	Weekly schedule definitions with per-day transition times and values.
Notif. Classes	Notification class instances with acknowledgement flags and recipient lists.

Trend Logs	Trend log instances with source reference, logging type, interval, and buffer size.
Network	BACnet/IP bridge port settings, secure-connect port, and global routing flag.
Remote Devices	Persisted remote device instances with IP address and network number.
Policies	Read/write policies — polling interval and max write time.
Raw JSON	Full raw JSON source.

7.2 network.json Report Tabs

Tab	Content
Overview	Hostname, config version, interface count, and routing priority.
Interfaces	Summary table of all interfaces — enabled state, IPv4 address, gateway, DHCP.
Bridge / Primary / Secondary / Auxiliary / Wireless	Detailed per-interface view: IPv4, IPv6, VLAN, and (for wireless) SSID and encryption.
Services	Spanning Tree Protocol mode and priority.
Routing	Interface priority order with enabled state and address.
Raw JSON	Full raw JSON source.

7.3 modbus.json Report Tabs

Tab	Content
Server	Modbus server enable state, address type, encoding / byte-swap settings, and server-side points.
<Device Name>	One tab per configured client device — connection details, encoding, and all point definitions grouped by register type (Coils, Discrete Inputs, Input Registers, Holding Registers).
Networks	Configured Modbus network entries.
Raw JSON	Full raw JSON source.

7.4 security.json Report Tabs

Tab	Content
Users	Per-user card showing roles, last login, last failed login, password creation date, reset-password flag, and mandatory account flag.
Policies	Lockout policy — max failed attempts and lockout duration. Authentication settings if present.
Raw JSON	Full raw JSON source.

7.5 iot.json Report Tabs

Tab	Content
Connections	Per-connection card showing provider, IoT Hub hostname, device ID, and redacted shared access key.
Raw JSON	Full raw JSON source.

Security note — IoT secrets
Shared Access Keys and other connection secrets found in iot.json are automatically redacted in the report view. They are never displayed in plain text.

8. Workspace Layout

The workspace folder is created automatically on first use. Its structure is:

Folder / File	Purpose
ExtractedFiles\	Full extraction target. Mirrors the internal ZIP folder tree exactly.
Zips\	Output ZIPs — DesignerProjects.zip and Dashboard.zip are saved here.
Logs\	Timestamped diagnostic log files (DCAFZipExtractor_<timestamp>.log).
DesignerProjects\	Legacy fallback folder for the designer ZIP source (if used manually).
Dashboards\	Legacy fallback folder for the dashboard ZIP source (if used manually).
report_bacnet.html	Last rendered BACnet report (written to workspace root on preview).
report_network.html	Last rendered Network report.
report_modbus.html	Last rendered Modbus report.
report_security.html	Last rendered Security report.
report_iot.html	Last rendered IoT report.

Default workspace location
If no workspace is specified, the tool creates a folder named `DCAF_Extract_<timestamp>` inside your system's Temp directory (`%TEMP%`). Use Change... to redirect to a persistent location if needed.

9. Diagnostic Log

Every operation is recorded with a timestamp and severity level. The log panel at the bottom of the window is collapsed by default; click the header to expand it.

Level	Meaning
INFO	Normal operational messages — file paths, scan results, extraction counts.
WARN	Non-fatal issues — Zip Slip attempts blocked, preview fallbacks triggered.
ERROR	Failed operations — extraction errors, report render failures.
DEBUG	Verbose detail — individual entry paths, stream operations.

Click **Save Log...** to export the current log to a file. Attach the log when raising a support issue with the tool itself.

10. Security & Safety

10.1 Read-Only ZIP Access

The source ZIP is always opened with read-only file access. The tool cannot modify, corrupt, or delete the original backup file.

10.2 Zip Slip Protection

Before any file is written during extraction, the tool verifies that the resolved destination path is inside the workspace's `ExtractedFiles` folder. Entries with absolute paths, drive letter prefixes, or path traversal sequences (`..`) are logged as warnings and skipped entirely.

10.3 Secret Redaction

IoT connection strings found in iot.json are parsed and their `SharedAccessKey` component is replaced with a *[redacted]* placeholder in the report view. The raw JSON tab shows the original file for authorised review.

11. Troubleshooting

Symptom	Resolution
Scan shows all artifacts as Missing	Verify the ZIP is a genuine Eclypse backup and not a different archive type. Check that the internal path prefix begins with bundle-content/.
Preview pane is blank after clicking Preview Selected	The artifact may be binary. Select Extract first, then use Open Selected.
Rich report renders but shows no data	The JSON file is empty or uses an unexpected structure. Switch to the Raw JSON tab to inspect the source.
Create DesignerProjects.zip shows a warning about no files	Run Extract first. The designer tree must be present in ExtractedFiles before re-zipping.

Open Selected opens nothing	The artifact has not been extracted yet. Run Extract and try again.
Application appears frozen during extraction	Large ZIPs with many files can take several seconds. The progress bar is active; wait for it to complete.
Zip Slip warning in the log	The backup ZIP contains a suspicious entry with a path traversal sequence. That entry is skipped. Report the ZIP to your security team.

12. Glossary

Term	Definition
Artifact	A specific file or folder tree expected inside an Eclypse backup ZIP.
BACnet	Building Automation and Control Networks — the communication protocol used by Eclypse controllers.
Eclypse	Distech Controls line of advanced building automation controllers.
EC-BOS	Eclypse Building Operating System — the firmware environment on Eclypse controllers.
Workspace	The local folder where extracted files, logs, and output ZIPs are stored.
Zip Slip	A class of archive vulnerability where crafted path entries escape the intended extraction directory.
STA (Single-Threaded Apartment)	A Windows threading model required by WPF user interfaces.